

Set 8 - Conditional Statements (if-else, switch, ternary)

Q1.

Write a Java program called FoodieBalanceChecker that takes an integer input for wallet balance and prints 'Low Balance Warning.' if the balance is less than 100, otherwise prints 'Sufficient Balance.' using an if-else statement.

Answer:

```
import java.util.Scanner;

public class FoodieBalanceChecker {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter your wallet balance: ");
        int balance = sc.nextInt();

        if (balance < 100) {
            System.out.println("Low Balance Warning.");
        } else {
            System.out.println("Sufficient Balance.");
        }

        sc.close();
    }
}

/* Sample Run:
Enter your wallet balance: 45
Low Balance Warning.
*/
```

Q2.

Create a Java class MovieAgeGate that asks the user for their age and uses an if-else-if ladder to print 'Kids Section' if age < 13, 'Teen Section' if age is between 13 and 17, and 'Adult Section' if age is 18 or above. Hint: Use Scanner for input and nested if-else for decision flow.

Answer:

```
import java.util.Scanner;

public class MovieAgeGate {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter your age: ");
        int age = sc.nextInt();

        if (age < 13) {
            System.out.println("Kids Section");
        } else if (age >= 13 && age <= 17) {
            System.out.println("Teen Section");
        } else {
            System.out.println("Adult Section");
        }

        sc.close();
    }
}
```

```

    }
}

/* Sample Run:
Enter your age: 15
Teen Section
*/

```

Q3.

Build a Java program called ZomatoCuisineFilter that takes a string input for cuisine ('Indian', 'Chinese', 'Italian', 'Mexican') and uses a switch statement to print a list of 2-3 sample restaurants for the selected cuisine. Print 'Cuisine Not Found' for any other input.

Answer:

```

import java.util.Scanner;

public class ZomatoCuisineFilter {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter cuisine (Indian/Chinese/Italian/Mexican): ");
        String cuisine = sc.nextLine();

        switch (cuisine) {
            case "Indian":
                System.out.println("Restaurants: Punjabi Tadka, Spice Villa, Saffron House");
                break;
            case "Chinese":
                System.out.println("Restaurants: Mainland China, Wok This Way, Chopstix");
                break;
            case "Italian":
                System.out.println("Restaurants: Pizza Express, La Pino'z, Olive Bistro");
                break;
            case "Mexican":
                System.out.println("Restaurants: Taco Bell, Chili's, El Sombrero");
                break;
            default:
                System.out.println("Cuisine Not Found");
        }

        sc.close();
    }
}

/* Sample Run:
Enter cuisine (Indian/Chinese/Italian/Mexican): Italian
Restaurants: Pizza Express, La Pino'z, Olive Bistro
*/

```

Q4.

Given this Java snippet: `int cartTotal = 450; String status = (cartTotal >= 500) ? "Eligible for Free Delivery" : "Add more items for Free Delivery";` Use ChatGPT to generate two alternate ways to write this logic using if-else and nested conditions. Copy and implement both in your code, then comment on which one you find clearer.

Answer:

```

public class FreeDeliveryCheck {

```

```

public static void main(String[] args) {
    int cartTotal = 450;

    // Original: ternary operator
    String status1 = (cartTotal >= 500) ? "Eligible for Free Delivery" : "Add more items for Free Delivery";
    System.out.println("Ternary version: " + status1);

    // Alternate 1: simple if-else
    String status2;
    if (cartTotal >= 500) {
        status2 = "Eligible for Free Delivery";
    } else {
        status2 = "Add more items for Free Delivery";
    }
    System.out.println("If-else version: " + status2);

    // Alternate 2: nested condition style
    String status3;
    if (cartTotal >= 500) {
        status3 = "Eligible for Free Delivery";
    } else {
        if (cartTotal < 500) {
            status3 = "Add more items for Free Delivery";
        } else {
            status3 = "Unknown";
        }
    }
    System.out.println("Nested-condition version: " + status3);
}

/* Sample Output:
Ternary version: Add more items for Free Delivery
If-else version: Add more items for Free Delivery
Nested-condition version: Add more items for Free Delivery
*/

```

Comment: The simple if-else version is the clearest for this case. The ternary operator is fine for a single quick condition but gets harder to read as logic grows, and the nested-condition version adds an unnecessary extra branch that never really gets used here (since the condition is a simple binary check). For more complex, multi-branch logic, if-else-if ladders or switch statements read better than deeply nested ternaries or nested ifs.

Q5.

Write a Java function `filterByMinRating(int[] ratings, int minRating)` that prints only those ratings from the array that are greater than or equal to `minRating`, similar to how Flipkart filters products by rating using SQL WHERE clauses. Hint: Use a for loop with an if condition to filter and print matching ratings.

Answer:

```

public class RatingFilter {

    public static void filterByMinRating(int[] ratings, int minRating) {
        System.out.println("Ratings >= " + minRating + ":");
        for (int i = 0; i < ratings.length; i++) {
            if (ratings[i] >= minRating) {
                System.out.println(ratings[i]);
            }
        }
    }
}

```

```
    }

    public static void main(String[] args) {
        int[] productRatings = {3, 5, 2, 4, 1, 5, 4, 3};
        filterByMinRating(productRatings, 4);
    }
}

/* Sample Output:
Ratings >= 4:
5
4
5
4
*/
```